

Generation Interconnection Request System Impact Study Report

For Queue Project Q11-0022

Revere 100 kV
June 24, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0022. The Interconnection Customer proposes a 607.3 MW natural gas combined cycle facility interconnecting at the Revere 100 kV substation in the Indiana area. The request was received on March 29, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 18 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$45,474,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Revere-E Astoria 100kV is overloaded to 585.2% of its 102 MVA emergency rating for the loss of the Calvert-Revee 100kV. The project contributes 100.0% DFAX (607.3 MW) to this overload.

The Revere-Calvert Substation 100kV Line is overloaded to 540.3% of its 111 MVA emergency rating for the loss of the E Astoria-Revere 100kV L. The project contributes 100.0% DFAX (607.3 MW) to this overload.

The Dalreed-Kemper County 100kV is overloaded to 338.0% of its 117 MVA emergency rating for the loss of the Drewry-Gaston Rd 100kV. The project contributes 72.7% DFAX (441.3 MW) to this overload.

For the loss of Revere-Calvert 100kV L, the East Astoria-Dalreed 100kV loads to 327.0% of its 123 MVA rating. Project DFAX: 65.5%.

For the loss of Revere-E Astoria 100kV, the Clvert-Dalreed 100kV loads to 289.7% of its 113 MVA rating. Project DFAX: 54.2%.

For the loss of Dalreed-Kemper County 100kV Line, the Gaston Rd Substation-Table Mt 100kV loads to 257.2% of its 138 MVA rating. Project DFAX: 67.1%.

The Gaston Rd-Drewry Substation 100kV L is overloaded to 257.0% of its 138 MVA emergency rating for the loss of the Kemper County-Dalreed 100kV. The project contributes 67.1% DFAX (407.4 MW) to this overload.

Colfax-E Astoria 100kV: 182.1% loading (136 MVA) following the outage of the Kemper County-Dalreed 100kV; DFAX = 48.5%.

The Calvert-Grandview Ave 100kV Line is overloaded to 178.8% of its 141 MVA emergency rating for the loss of the Kemper County-Dalreed 100kV. The project contributes 51.5% DFAX (313.1 MW) to this overload.

S Bend Substation-Kemper County 100kV: 168.6% loading (96 MVA) following the outage of the Drewry-Gaston Rd 100kV; DFAX = 28.0%.

For the loss of E Astoria Substation-Revere 100kV, the Drewry-Grandview Ave 100kV loads to 152.7% of its 118 MVA rating. Project DFAX: 25.7%.

For the loss of Kemper County-Dalreed 100kV, the Colfax-Drewry 100kV Line loads to 136.8% of its 129 MVA rating. Project DFAX: 34.5%.

The Table Mt-Suter 100kV L is overloaded to 131.3% of its 144 MVA emergency rating for the loss of the Kemper County-Dalreed 100kV. The project contributes 24.6% DFAX (149.6 MW) to this overload.

Suter-High Falls - Croghan Ny 100kV: 120.3% loading (138 MVA) following the outage of the Dalreed-Kemper County 100kV L; DFAX = 24.6%.

The S Bend-High Falls - Croghan Ny 100kV is overloaded to 110.3% of its 121 MVA emergency rating for the loss of the Drewry-Gaston Rd 100kV Line. The project contributes 28.0% DFAX (170.3 MW) to this overload.

The Ager-Northern 100kV line is overloaded to 109.9% of its 128 MVA emergency rating for the loss of the Kemper County-Dalreed 100kV Line. The project contributes 25.4% DFAX (154.1 MW) to this overload.

For the loss of Kemper County-S Bend 100kV, the Twin Branch-Kemper County 100kV Line loads to 104.3% of its 135 MVA rating. Project DFAX: 34.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Revere-East Astoria 100kV	Rebuild line with higher-capacity conductor	\$3,340,000	100% (\$3,340,000)
Revere-Calvert 100kV line	Rebuild line with higher-capacity conductor	\$1,838,000	100% (\$1,838,000)
Kemper County-Dalreed 100kV	Rebuild line with higher-capacity conductor	\$1,107,000	100% (\$1,107,000)
East Astoria-Dalreed 100kV	Rebuild line with higher-capacity conductor	\$1,600,000	100% (\$1,600,000)
Calvert-Dalreed 100kV Line	Rebuild line with higher-capacity conductor	\$7,105,000	100% (\$7,105,000)
Table Mt-Gaston Rd 100kV L	Rebuild line with higher-capacity conductor	\$998,000	100% (\$998,000)
Gaston Rd-Drewry 100kV line	Rebuild line with higher-capacity conductor	\$2,541,000	100% (\$2,541,000)
Colfax-East Astoria 100kV	Rebuild line with higher-capacity conductor	\$2,577,000	100% (\$2,577,000)
Calvert-Grandview Ave 100kV Line	Rebuild line with higher-capacity conductor	\$3,621,000	100% (\$3,621,000)
S Bend Substation-Kemper County 100kV	Rebuild line with higher-capacity conductor	\$2,163,000	100% (\$2,163,000)
Drewry-Grandview Ave 100kV	Rebuild line with higher-capacity conductor	\$2,949,000	100% (\$2,949,000)
Colfax-Drewry 100kV Line	Rebuild line with higher-capacity conductor	\$1,896,000	100% (\$1,896,000)
Suter-Table Mt 100kV	Rebuild line with higher-capacity conductor	\$7,205,000	100% (\$7,205,000)
Suter-High Falls - Croghan Ny 100kV	Rebuild line with higher-capacity conductor	\$2,528,000	100% (\$2,528,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
High Falls - Croghan Ny-South Bend 100kV	Rebuild line with higher-capacity conductor	\$1,463,000	100% (\$1,463,000)
Ager-Northern 100kV line	Reconductor line	\$1,073,000	100% (\$1,073,000)
Twin Branch-Kemper County 100kV	Reconductor line	\$1,470,000	100% (\$1,470,000)
	Total allocated to project		\$45,474,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.